RESUME

Gregor Nottmeier

M.Sc. Data Science Student | Applied AI & Reinforcement Learning
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Languages: German (Native/C2), English (Fluent/C1), Slovak (Fluent/C1), Spanish (Basic/A1-A2), Latin (Latinum)

PROFESSIONAL SUMMARY

M.Sc. Data Science student at the University of Mannheim with a strong mathematical foundation (B.Sc. Mathematics in Business and Economics) and hands-on experience in reinforcement learning, statistical modeling, and full-stack ML architectures. Proven track record in developing modular data pipelines, fine-tuning transformer models, and automating analytics workflows. Eager to leverage strong algorithm design and object-oriented programming skills to drive next-generation supply chain simulation and industrial optimization at adago.

TECH-STACK & SKILLS

- **Programming Languages:** Python (OOP), SQL, TypeScript, MATLAB, VBA, SAS, LaTeX
- **Machine Learning & Frameworks:** PyTorch, Hugging Face Transformers, Stable-Baselines3, Scikit-Learn, Gymnasium, Braindecode, GeoPandas, OSMnx
- **Backend & Infrastructure:** Docker, PostgreSQL, Flask, Next.js, Nginx, Gunicorn, systemd, Prisma, Linux, Git, GitLab, GitHub, Jupyter
- **Core Competencies:** Reinforcement Learning & Bandit Algorithms, Deep Learning, Large Language Models (LLMs), Statistical Modeling, Time Series Analysis, Simulation & Optimization, Full-Stack ML Deployment

SELECTED PROJECTS

LLM Sentiment Alignment via RLHF
Tilburg University | Python, PyTorch, Transformers, Stable-Baselines3, Gymnasium

- Enhanced model alignment stability and mitigated reward-hacking by systematically tuning KL-divergence penalties and adaptive batch sizes during GPT-2 fine-tuning.
- Improved positive sentiment generation scores while preserving linguistic diversity by architecting an end-to-end Deep Reinforcement Learning pipeline.

Interactive Surveys Web-App & ML Analysis Pipeline
Team Project (M.Sc. Data Science) | Flask, PostgreSQL, Nginx, Gunicorn, systemd, GitLab

- Built a high-performance research platform for analyzing Human-AI interaction dynamics by engineering an asynchronous Flask backend connected to raw PostgreSQL database layers.
- Streamlined downstream machine learning evaluation by orchestrating modular background subprocesses and deploying robust production infrastructure under Linux and Nginx.

Contextual Linear Bandit Reinforcement Learning (linUCB / OFUL)
Bachelor Thesis (Universität Mannheim) | Python, Reinforcement Learning, Monte Carlo Methods

- Evaluated theoretical regret bounds and algorithm efficiency across dynamic multi-armed bandit settings by implementing custom linUCB and OFUL simulation environments in Python.
- Achieved top academic distinction (Grade: 1.3) by rigorously bridging mathematical optimization theory with stochastic empirical modeling.

PROFESSIONAL EXPERIENCE

Mercedes-Benz Bank AG — Intern Risk Analytics Credit
10.2024 – 02.2025 | SAS, Excel (VBA), SAP Bank Analyzer, Power BI

- Accelerated month-end closing processes and reduced manual execution time by developing automated VBA macros and SAS data aggregation workflows.
- Enhanced credit risk monitoring by designing interactive Power BI dashboards to track Cost of Risk metrics and portfolio exposure.

Integra e.V. (Student Consulting) — Internal Finance Manager & Project Lead
03.2021 – 02.2023 | Project Management, MS Office, Market Analysis

- Delivered strategic insights for enterprise clients by co-directing comprehensive B2B customer experience market research initiatives.
- Ensured organizational financial stability and operational alignment by managing budget allocation, workshops, and internal strategy projects.

C4 Steuerstrategen GmbH — Working Student
07.2022 – 09.2022 | Data Management, Process Optimization

- Maintained 100% data compliance and audit readiness by structuring, verifying, and preprocessing real estate tax assessment records.

Lerne Smart UG / J.C.C. Bruns — Content Creator & Digital Educator
02.2019 – 09.2020 | Digital Education, Agile Execution

- Increased digital literacy across target groups by designing and delivering structured, interactive training modules for mobile and desktop systems.

EDUCATION

Universität Mannheim — M.Sc. Data Science
09.2024 – Present | Current GPA: 1.9 (German Scale)

- Focus: Machine Learning, Reinforcement Learning, Natural Language Processing, Distributed Computing.

Tilburg University — Erasmus+ Exchange Semester (Data Science / AI)
01.2026 – 06.2026 | Grade: 8.6 / 10 (Excellent)

- Focus: Advanced Deep Reinforcement Learning, LLM Alignment, NLP Architectures.

Universität Mannheim — B.Sc. Mathematics in Business and Economics
09.2020 – 09.2024 | Final Grade: 2.2 | Bachelor Thesis Grade: 1.3

- Thesis: The linUCB / OFUL Algorithm in Contextual Linear Bandit Reinforcement Learning.

Universidad Autónoma de Madrid — Erasmus+ Exchange Semester
09.2023 – 01.2024 | Mathematics & Economic Modeling

Ratsgymnasium Minden — Allgemeine Hochschulreife (Abitur)
Graduated 06.2020 | Final Grade: 1.2

VOLUNTEERING & ENGAGEMENT

- **Member of Student Council (Fachschaft Math/Info), Universität Mannheim** (Until 10.2024): Represented student body interests at faculty board meetings and organized technical networking events.
- **Volunteer Enumerator / Field Interviewer, Zensus 2022 (City of Mannheim)** (05.2022 – 10.2022): Executed structured household survey interviews adhering to strict data quality and privacy standards.